Problem Statement

The foundation of the Indian economy is agriculture. The most crucial factor for agriculture is the water source, or rainfall. Farmers are alerted by the rainfall prediction because they can safeguard their crops from the rain early on. Therefore, it is critical to make as accurate of a rainfall prediction as possible. Investigation and analysis of rainfall data across different locations in India, with a focus on those where agricultural activity has been done consistently across a wide range.
Future rainfall prediction for certain regions utilising various machine learning approaches, such as XGBoost classifier, SVM classifiers, Decision tree, Naive Bayes classifier, Logistic regression, etc., with the aid of the analysis of the resulting data.

Vignesh M

- Make sure the data are accurate and sufficient for predictions.
- Utilizing data from the Indian Meteorological Department to track wind direction and speed, and development of cyclones in the Arabian Sea and the Bay of Bengal
- Daily observation of the clouds' density in the sky.
- Examining the atmospheric pressure in several, physically distinct places.

Vikneshwar A

- Gathering timely temperature data from different areas and subdivisions.
- Naive Bayes Classifier
- Using logistic regression, forecast rainfall for tomorrow "Yes" or "No"
- Evaluating model performance by comparing our datasets and predictions with those from the past.
- XGBoost Classifier

Common Ideas shared

Random Forest Classifier	Using crop yields from the previous season, month, or year to analyse and predict rainfall.	To estimate rainfall (in mm) based on characteristics, use linear regression	K-Nearest Neighbours
Examining floods and encroachments to view the impact of rainfall	Time Series Forecasting		

Srivathassan V

- Observing and recording daily temperature data.
- Observing or analysing seasonal trends in rainfall.
- Monitoring wind direction and speed.
- Examining the areas or districts most likely to have rain.
- Neural Network Classifiers

Vignesh A

- SVM Classifier
- Precipitation forecasting based on data from prior years.
- Decision Tree
- Gathering data and discovering trends therein.

Data Mining Ideas

Precipitation forecasting based on data from prior years.	Daily observation of the clouds' density in the sky	Using crop yields from the previous season, month, or year to analyse and predict rainfall.	Examining floods and encroachments to observe the effects of rainfall.

Agricultural Intention

Gathering data and understanding the trends it contains.	Indian Metrological Department Data & News for Observing Wind Direction & Cyclone Formation in the Arabian Sea & Bay of Bengal	Examining the different, geographically distinct places' atmospheric pressure.	Gathering timely temperature data from different areas and subdivisions.	Observing or analysing seasonal rainfall trends.

Data Mining Ideas

AI Approaches

ML Algorithms

- XGBoost Classifier
- Naive Bayes Classifier
- Using logistic regression, forecast rainfall for tomorrow "Yes" or "No"
- K-Nearest Neighbours
- Decision Tree
- Time Series Forecasting
- Random Forest Classifier
- SVM Classifier
- To estimate rainfall (in mm) based on characteristics, use linear regression

Deep Learning Techniques

- Neural Network Classifiers